

Problem Set 1

Applied Stats/Quant Methods 1

Due: October 8, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 8, 2026. No late assignments will be accepted.

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```
1 #####
```

Number of Observations::

```
1 # a.
```

Visualising the Data:

```
1 # hypothesis test
2 # variables:
3 # explanatory: students (numeric)
4 # response: IQ scores (numeric)
5
6 # i. assumptions:
7 # - numeric, continuous data
8 # - sample size = 25
9 # normal distribution
```

```

10
11 # visualizing the distribution: we graph the observations to ensure for
    normality since the sample size is less than 30. We observe a
    relatively normal distribution , hence we can proceed with the
    significance test .
12 pdf("plot_1.pdf")

```

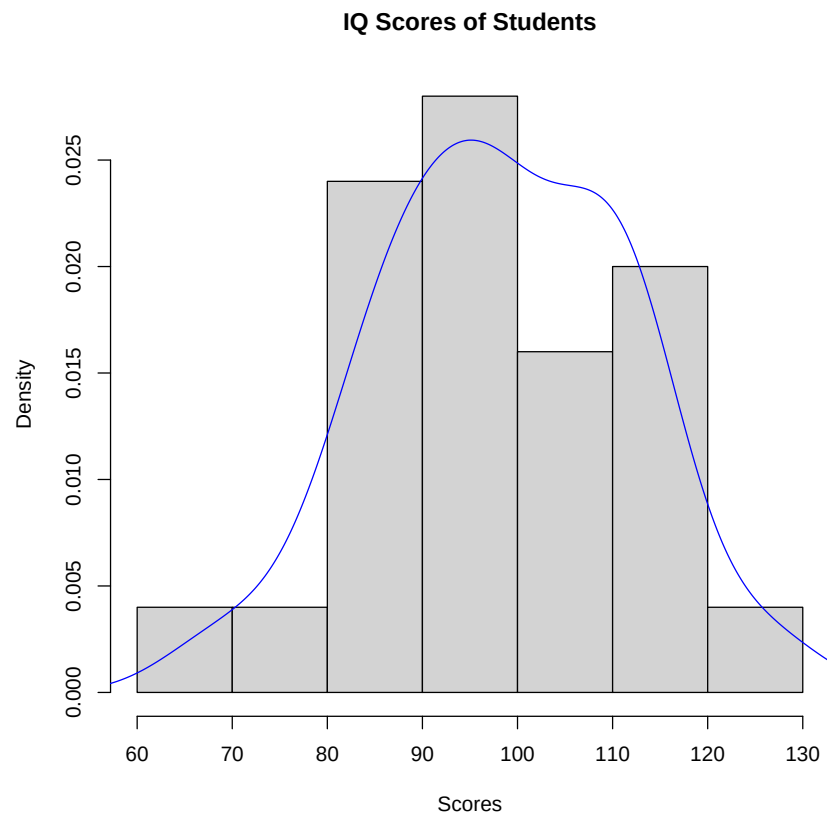


Figure 1: IQ Scores of Students

1. Find a 90% confidence interval for the average student IQ in the school (As a hint, you first need the mean and SD to create a CI):

Descriptive Statistics:

```

1
2 # descriptive statistics
3 mean_y <- sum(y)/n

```

```

4 sd_y <- sqrt(sum((y - mean_y) ^ 2) / (n - 1))
5 se_y <- sd_y/sqrt(n)
6
7 # calculating the t-critical value: we use the t-distribution instead of
  the z-distribution because we observe that the length of y is less
  than 30 and the population mean and standard deviations are both
  unknown.

```

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

State	50 states in US
Y	per capita expenditure on shelters/housing assistance in state
X1	per capita personal income in state
X2	Number of residents per 100,000 that are "financially insecure" in state
X3	Number of people per thousand residing in urban areas in state
Region	1=Northeast, 2= North Central, 3= South, 4=West

Explore the `expenditure` data set and import data into R.

```
1 # HA: average IQ score of the students is greater than 100      (mu > 100)
```

- Please plot the relationships among Y , $X1$, $X2$, and $X3$? What are the correlations among them (you just need to describe the graph and the relationships among them)?

```
1 # We fail to reject the null because there is not sufficient evidence to
  say that the average IQ score in the counselor's school is higher than
  the average IQ score among all schools in the country.
2 # We draw this conclusion from observing a p-value = 0.7215, which is
  significantly greater than the alpha threshold value = 0.05.
3 # In addition, we also observe that the t-score of -0.596 exceed the
  upper tail t-score of 1.711.
4 # Hence, in a world where the null is true, it is unlikely that we
  observe a value more extreme than the observed value.
5
6 # checking my work with the built-in t.test function
7 t.test(y, mu = 100, alternative = "greater", conf.level = 0.95)
8
9 #####
10 # Problem 2
11 #####
12
13 expenditure <- read.table("https://raw.githubusercontent.com/ASDS-TCD/
  StatsI_2026/main/datasets/expenditure.txt", header=T)
14
15 # quick overview
16 head(expenditure)
17 str(expenditure)
18 summary(expenditure)
19
20 # 1. Question 2: Political Economy
```

```

21
22 # a.
23
24 vars <- c("Y", "X1", "X2", "X3")
25
26 labels <- c(
27   "Expenditure",
28   "Income",
29   "Financially Insecure",
30   "Urban"
31 )
32
33 # use scatterplots for each panel because it most accurately represents
34   the correlation between two continuous variables, displaying the
35   magnitude and direction with the line of best fit
36 # scatterplots also efficiently visualise any outliers that may be
37   skewing the data
38 pdf("plot_2.pdf", width = 12, height = 8)
39 par(mfrow = c(2, 3))
40
41 # use for loop to create the same six graphs with its respective
42   variables
43 # abline creates a line of best fit
44 # mtext displays the correlation r value
45 for (i in 1:(length(vars) - 1)) {
46   for (j in (i + 1):length(vars)) {

```

- Please plot the relationship between Y and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

```

1   expenditure[[vars[i]]],
2   xlab = labels[j],
3   ylab = labels[i],
4   main = paste(labels[i], "vs", labels[j]),
5   cex = 0.4
6 )
7
8 abline(
9   lm(expenditure[[vars[i]]] ~ expenditure[[vars[j]]]),
10  col = "blue",
11 )
12
13 r <- cor(
14   expenditure[[vars[i]]],
15   expenditure[[vars[j]]]
16 )
17
18 mtext(
19   paste("r =", round(r, 2)),
20

```

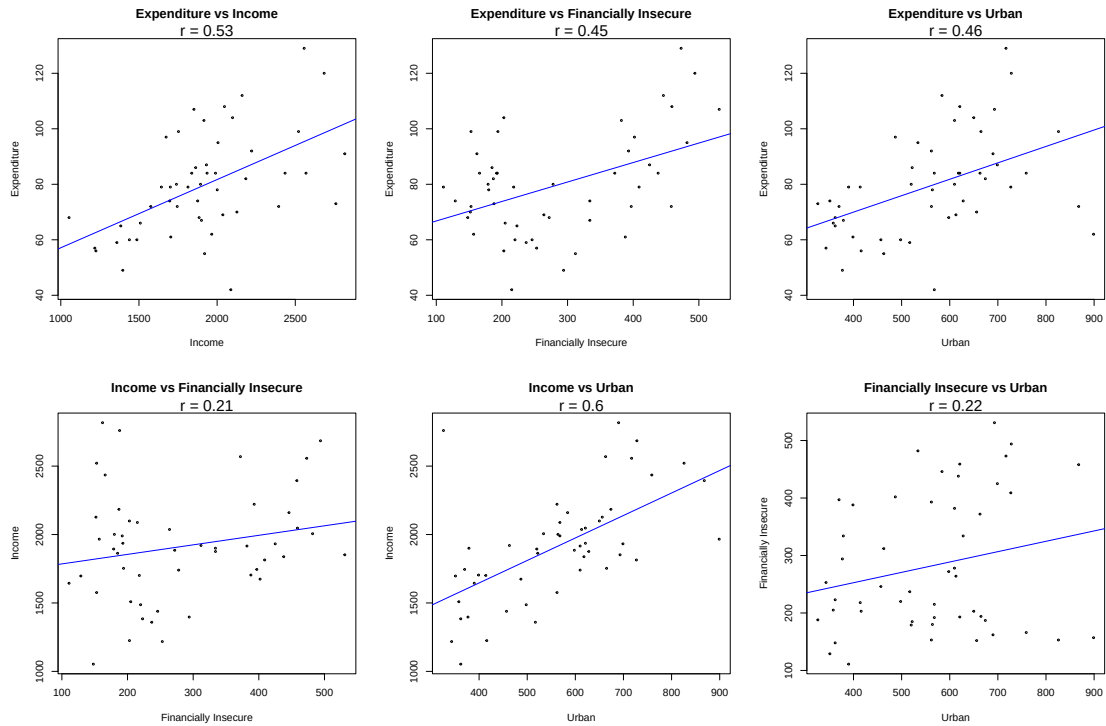


Figure 2: Pairwise Relationships Between Expenditure Variables

21

- Please plot the relationship between Y and $X1$? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

```

1 dev.off()
2
3 # expenditure vs income (0.53): positive moderate linear relationship
4 # expenditure vs financially insecure (0.45): positive moderate linear
  relationship
5 # expenditure vs urban (0.46): positive moderate linear relationship
6 # income vs financially insecure (0.21): positive weak linear
  relationship
7 # income vs urban (0.6): positive strong linear relationship
8 # financially insecure vs urban (0.22): positive weak linear relationship

1 # Income and individuals who reside in urban areas have the strongest
  positive linear relationship .
2 # "Financially insecure" residents has a weak positive relationship with
  income and individuals who reside in urban areas, since in both graphs
  , the observations are widely around the fitted line.
3 # In general, higher per capita income, levels of financial insecurity,
  and urban population are each associated with higher per capita

```

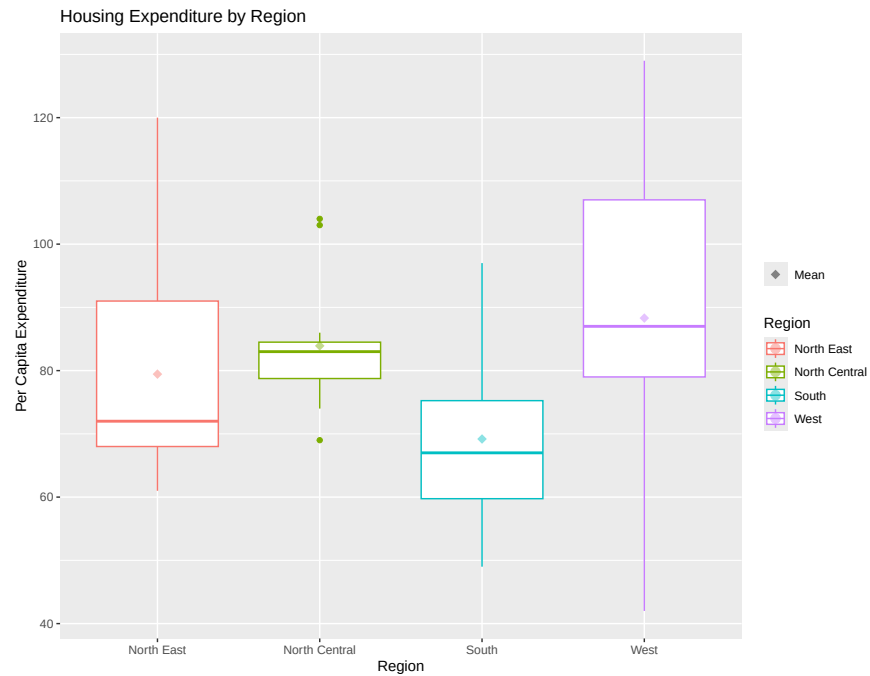


Figure 3: Housing Expenditure by Region



Figure 4: Housing Expenditure versus Income

housing expenditure.

```

4
5 # b.
6
7 # use factor() to separate housing expenditure by region
8 # labeling the four different regions to their respective names: 1 =
    northeast, 2 = north central, 3 = south, 4 = west
9 regions <- factor(
10 expenditure$Region,
11 levels = c(1, 2, 3, 4),
12 labels = c("North East", "North Central", "South", "West")
13 )
14
15 # comparing that the mean and median are relatively similar, so that a
    boxplot would be a good representation of comparing the average
    housing expenditure by region
16 # in each region, the mean and median have a negligible difference
17 for (i in levels(regions)) {
18   region_mean <- round(mean(expenditure$Y[regions == i]), 2)
19   region_median <- round(median(expenditure$Y[regions == i]), 2)

```



Figure 5: Housing Expenditure versus Income by Region